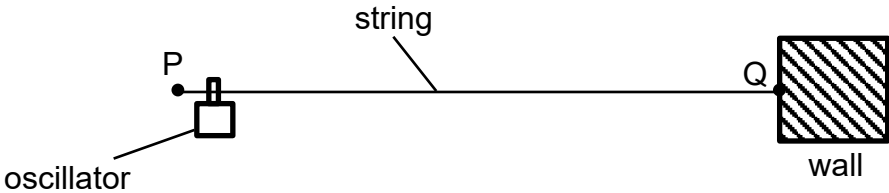
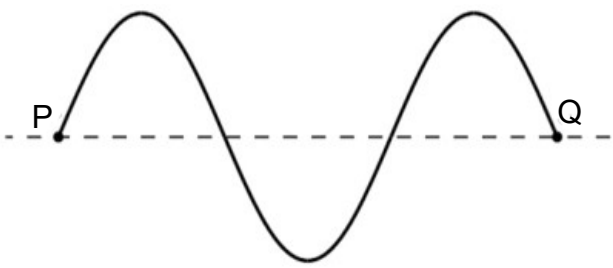
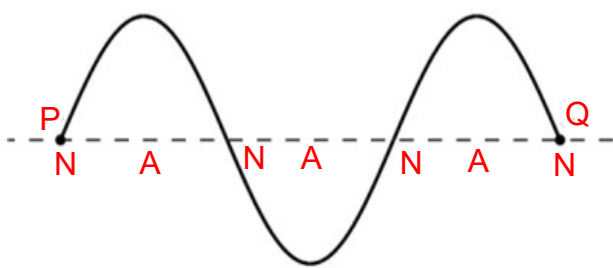
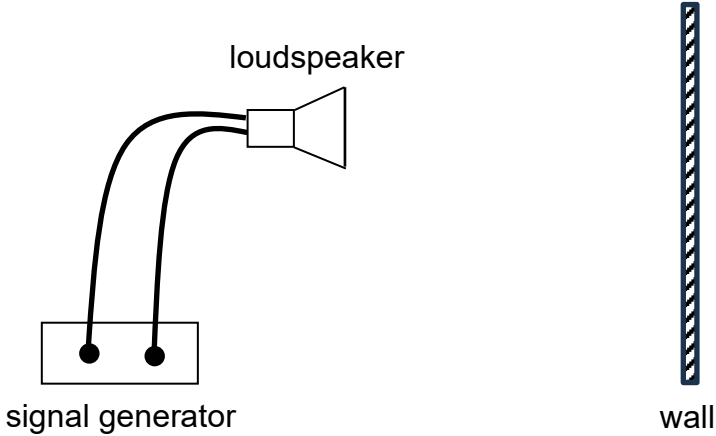


3	(a)	Fig. 3.1 shows a string stretched between two fixed points P and Q.  Fig. 3.1 An oscillator is attached near end P of the string. End Q is fixed to a wall. The oscillator has a frequency of 480.0 Hz. The stationary wave produced on PQ at an instant time t is shown in Fig. 3.2. Each point on the string is at its maximum displacement.  Fig. 3.2
	(i)	On Fig. 3.2, label all the nodes with the letter N and the antinodes with the letter A along the dotted line PQ.
		[2]
		Solution  Correct position of nodes drawn. Correct position of antinodes drawn. B1 B1
	(ii)	Calculate the lowest possible frequency of the wave that can be formed between end P and Q of the string.
		frequency = Hz [3]

			Solution Using $v = f\lambda$, therefore $f = \frac{v}{\lambda}$. There are 1.5 wavelengths of wave formed between end P and Q for the string in Fig. 3.2. Length L of string between end P and Q: $L = \frac{3\lambda_3}{2}$ Frequency in Fig. 3.2: $f = \frac{v}{\lambda_3} = \frac{v}{\frac{2L}{3}} = \frac{3v}{2L} = 480.0 \text{ Hz}$ For lowest possible frequency formed between end P and Q of string, it corresponds to 0.5 wavelength. Length L of string between end P and Q: $L = \frac{\lambda_1}{2}$ Lowest possible frequency: $f' = \frac{v}{\lambda_1} = \frac{v}{\frac{2L}{3}} = \frac{3v}{2L} \left(\frac{1}{3} \right) = \frac{480.0}{3}$ $= 160.0 \text{ Hz}$	C1	M1	A1
	(b)	A loudspeaker is connected to a signal generator. It is then oriented to face a wall as shown in Fig. 3.3. Sound waves are produced between the speaker and the wall to form a stationary wave.  loudspeaker signal generator wall Fig. 3.3				
	(i)	Explain why there are alternate regions of high and low intensity detected between the loudspeaker and the wall. 				
						[3]
			Solution The sound waves propagate and <u>reflect off the wall, the incident and reflected waves have the same amplitude, wavelength/ frequency/ speed,</u>	B1		

			<u>They move in opposite direction and overlap/superpose/meet.</u>	B1
			When they experience <u>constructive interference</u>, they form <u>regions of high intensity</u> and when they experience <u>destructive interference</u>, they form <u>regions of low intensity</u>.	B1
		(ii)	Outline how you would use this setup to obtain the speed of sound waves in air.	
				
				
				
				
				
				
				
				
				
				
				[4]
			Solution A <u>microphone</u> is moved between the speaker and the wall to detect maximum and minimum intensities. The microphone is connected to a cathode ray oscilloscope (CRO). The frequency is calculated from the period obtained from the CRO. Measure the <u>separation between the positions of adjacent minimum/maximum intensities</u> which is equivalent to half a wavelength. Repeat the measurements of the distance between these positions and find the average. <u>Multiply the average inter-nodal distance</u> to obtain the wavelength of the wave. Use <u>$v = f\lambda$</u> to determine the speed of sound waves in air.	B1 B1 B1 B1

[Total: 12]

